



RMC Homework-1

BY: KANDIRAJU VENKATA SAI ADVAITH
ROBOT MECHANICS AND CONTROL
ME 5250.

1.(a)

For a, b, c to be linearly independent,

$$a\theta_1 + b\theta_2 + c\theta_3 = 0$$

$$\theta_1 = \theta_2 = \theta_3 = 0$$

$$\Rightarrow a \begin{bmatrix} 4 \\ -9 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 13 \\ 10 \end{bmatrix} + c \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

which can be written as,

$$\begin{bmatrix} 4 & 2 & 2 \\ -9 & 13 & -4 \\ 1 & 10 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By applying gaussian elimination; $\theta_1, \theta_2, \theta_3$ can be obtained

$$\begin{bmatrix} 4 & 2 & 2 & 0 \\ -9 & -13 & -4 & 0 \\ 1 & 10 & 1 & 0 \end{bmatrix}$$

$$R_3 \leftrightarrow R_1$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ -9 & -13 & -4 & 0 \\ 4 & 2 & 2 & 0 \end{bmatrix}$$

$$\text{Now, } R_2 = R_2 + 9R_1$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 103 & 5 & 0 \\ 4 & 2 & 2 & 0 \end{bmatrix}$$

$$R_3 = R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 103 & 5 & 0 \\ 0 & -38 & -2 & 0 \end{bmatrix}$$

$$R_3 = R_3 / (-2)$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 103 & 5 & 0 \\ 0 & 19 & 1 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 19 & 1 & 0 \\ 0 & 103 & 5 & 0 \end{bmatrix}$$

$$R_2 = R_2 - R_1$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 103 & 5 & 0 \end{bmatrix}$$

$$R_2 = R_2 / 9$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 103 & 5 & 0 \end{bmatrix}$$

$$R_3 = R_3 - 103R_2$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix}$$

$$R_3 = R_3 / 5$$

$$\begin{bmatrix} 1 & 10 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R_1 = R_1 - R_3, \quad R_1 = R_1 - 10R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

From this equation, we can say that $\theta_1 = \theta_2 = \theta_3 = 0$.

$\therefore$ The given set of vectors are independent in that associated vector field.

1.(b)

$$a = \begin{bmatrix} 1+i \\ 2+3i \end{bmatrix} \quad b = \begin{bmatrix} 10+2i \\ 4-i \end{bmatrix} \quad c = \begin{bmatrix} -i \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1+i & 10+2i & -i \\ 2+3i & 4-i & 3 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_1 = 3R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 26+7i & -3-3i & 0 \\ 2+3i & -4-i & 3 & 0 \end{array} \right]$$

$$R_2 = R_2 - (2+3i)R_1$$

$$\left[\begin{array}{ccc|c} 1 & 26+7i & -3-3i & 0 \\ 0 & -27-93i & 15i & 0 \end{array} \right]$$

$$R_2 = R_2 / (9+3i)$$

$$\left[\begin{array}{ccc|c} 1 & 26+7i & -3-3i & 0 \\ 0 & 1 & -15/1042 - 45i/1042 & 0 \end{array} \right]$$

$$R_1 = R_1 - (26+7i)R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 589/1042 - 871i/1042 & 0 \\ 0 & 1 & -15/1042 - 45i/1042 & 0 \end{array} \right]$$

2) For vectors to be in subspace, it has to follow vector addition as well as scalar multiplication i.e both has to co-exist in the subspace.

(a) Let (b_1, b_2, b_3) & (c_1, c_2, c_3) exist in subspace A
 $b_1=0$ $c_1=0$

vector addition:

$$\begin{aligned} (b_1, b_2, b_3) + (c_1, c_2, c_3) &= (b_1+c_1, b_2+c_2, b_3+c_3) \\ &= (0, b_2+c_2, b_3+c_3) \end{aligned}$$

This exists in the subspace.

Scalar Multiplication:

$$c(b_1, b_2, b_3) = (cb_1, cb_2, cb_3) = (0, cb_2, cb_3)$$

which exists in the subspace.

$\therefore$ Plane of vectors with first component $x_1=0$ is a subspace.

(b) Let us say two vectors (b_1, b_2, b_3) and (c_1, c_2, c_3) where $b_1=1$ and $c_1=1$.

vector addition:

$$(1, b_2, b_3) + (1, c_2, c_3) = (2, b_2+c_2, b_3+c_3)$$

This doesn't exist in the subspace.

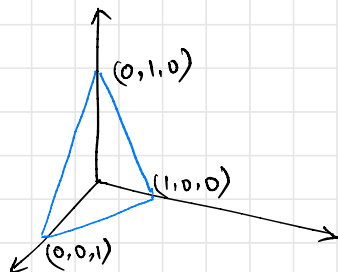
Scalar multiplication:

$$c(1, b_2, b_3) = (c, b_2, b_3)$$

This doesn't exist in subspace.

$\therefore$ plane of vectors with first component $x_1=1$ is not subspace.

- (c) Suppose that the plane passing through the end point of orthogonal basis vectors be represented as:



equation of plane:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\frac{x}{1} + \frac{y}{1} + \frac{z}{1} = 1$$

$(x, y, z) = (0, 0, 0)$ does not satisfy plane equation.

$\therefore$ plane goes through end point of orthogonal basis vectors is not a subspace.

- (d) $\alpha_1 (1, 1, 0), \beta_1 (2, 0, 1)$

Vector addition:

$$\begin{aligned} \alpha_1 (1, 1, 0) + \beta_1 (2, 0, 1) + \alpha_2 (1, 1, 0) + \beta_2 (2, 0, 1) \\ = (\alpha_1 + \alpha_2) (1, 1, 0) + (\beta_1 + \beta_2) (2, 0, 1) \end{aligned}$$

This exists in subspace.

Scalar Multiplication:

$$c\alpha_1 (1, 1, 0) + c\beta_1 (2, 0, 1)$$

which also exists in the subspace.

$\therefore$ All the combinations of $(1, 1, 0)$ & $(2, 0, 1)$ are a subspace.

(e) consider two vectors (b_1, b_2, b_3) and (c_1, c_2, c_3) satisfying $3x_1 - x_2 + x_3 = 0$

vector addition:

$$\begin{aligned}(b_1, b_2, b_3) + (c_1, c_2, c_3) &= (b_1 + c_1, b_2 + c_2, b_3 + c_3) \\ (3b_1 + 3c_1, -b_2 - c_2, b_3 + c_3) &= (3b_1, -b_2, b_3) + (3c_1, -c_2, c_3) \\ &= 0 + 0 = 0.\end{aligned}$$

This exists in subspace.

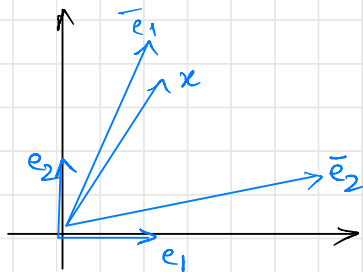
Scalar multiplication:

$$\alpha(b_1, b_2, b_3) = \alpha(3b_1, -b_2, b_3) = \alpha \cdot 0 \Rightarrow 0.$$

exists in subspace.

$\therefore$ **vectors satisfying $3x_1 - x_2 + x_3 = 0$ are a subspace.**

$$3. \quad x = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \bar{e}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \bar{e}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$



$$x = [e_1, e_2] \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\bar{e}_1 = [e_1, e_2] \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\bar{e}_2 = [e_1, e_2] \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Analytically: $x = a_1 e_1 + b_1 e_2$

$$(1, 3) = a_1(1, 0) + b_1(0, 1)$$

$$a_1 = 1, \quad b_1 = 3$$

$$x = a_2 \bar{e}_1 + b_2 \bar{e}_2$$

$$(1, 3) = a_2(1, 4) + b_2(3, 2)$$

$$a_2 + 3b_2 = 1 \quad 4a_2 + 2b_2 = 3$$

$$\Rightarrow a_2 = 7/10 \quad b_2 = 1/10.$$

Graphically;

$$x = [e_1 \ e_2] \begin{bmatrix} 1 \\ 3 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

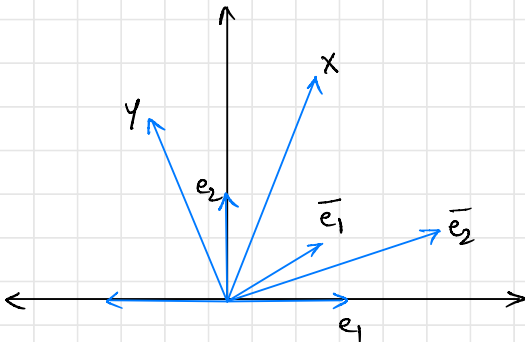
$$\bar{e}_1 = [e_1 \ e_2] \begin{bmatrix} 1 \\ 4 \end{bmatrix} \Rightarrow \bar{e}_2 = [e_1 \ e_2] \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$Q = P^{-1} = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \quad P = \frac{1}{10} \begin{bmatrix} -2 & 3 \\ 4 & -1 \end{bmatrix}$$

$$\bar{B} = PB = \frac{1}{10} \begin{bmatrix} -2 & 3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 7 \\ 11 \end{bmatrix}$$

$$x = [\bar{e}_1 \ \bar{e}_2] \begin{bmatrix} 7/10 \\ 11/10 \end{bmatrix}$$

4)



$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\bar{e}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \bar{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x = [e_1 \ e_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow P = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y_1 = [e_1 \ e_2] \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad y_2 = [e_1 \ e_2] \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\bar{e}_1 = [e_1 \ e_2] \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \bar{e}_2 = [e_1 \ e_2] \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Q = P^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\bar{B} = PB = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ -x_1 + 2x_2 \end{bmatrix}$$

$$\bar{A} = P A P^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

5) $L[x] = \begin{bmatrix} x_1 - x_2 \\ x_2 - x_3 \end{bmatrix} = y_2$

$$y = a_1 w_1 + a_2 w_2$$

$$= a_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$L \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} ; L \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} ; L \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

For v_1 :

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_{11} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_{21} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$a_{11} + a_{21} = 1 \quad \text{and} \quad a_{11} - a_{21} = 0$$

$$a_{11} = a_{21} \Rightarrow 2a_{11} = 1$$

$$\Rightarrow \underline{a_{11} = 0.5}$$

For v_2 :

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = a_{12} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_{22} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$a_{12} + a_{22} = 1 ; a_{12} - a_{22} = 1$$

$$2a_{12} = 0 \Rightarrow \underline{a_{12} = 0}$$

$$\text{i.e. } \underline{a_{22} = 1}$$

For V_3 :

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} = a_{13} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + a_{23} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$a_{13} + a_{23} = -1$$

$$a_{13} - a_{23} = 2$$

$$2a_{13} = 1 \Rightarrow a_{13} = 1/2, a_{23} = -3/2$$

representation of I :

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/2 & 1 & -3/2 \end{bmatrix}$$

7)

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -7 \end{bmatrix}$$

Augmented matrix,

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 4 & 1 & 0 & 2 \\ -2 & 2 & 1 & -7 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & -4 & -6 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 / -1 \\ R_1 \rightarrow R_1 / 2 \\ R_3 \rightarrow -R_3 / 4 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1/2 & 1/2 & 1/2 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3/2 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_3 \\ R_1 \rightarrow R_1 - 1/2 R_3 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1/2 & 0 & -1/4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 3/2 \end{array} \right]$$

$$R_1 \rightarrow R_1 - \frac{R_2}{2} \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5/4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 3/2 \end{array} \right]$$

Hence, $x_1 = 5/4$

$x_2 = -3$

$x_3 = 3/2$

$$7b) \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}$$

Augmented matrix

$$\begin{bmatrix} 1 & 3 & 3 & 2 & | & 1 \\ 2 & 6 & 9 & 5 & | & 5 \\ -1 & -3 & -3 & 0 & | & 5 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 3 & 3 & 2 & | & 1 \\ 0 & 0 & 3 & 1 & | & 3 \\ 0 & 0 & 6 & 2 & | & 6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2 \quad \begin{bmatrix} 1 & 3 & 3 & 2 & | & 1 \\ 0 & 0 & 3 & 1 & | & 3 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \quad \begin{bmatrix} 1 & 3 & 0 & 1 & | & -2 \\ 0 & 0 & 3 & 1 & | & 3 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_1 + 3x_2 + x_4 = -2 ;$$

$$3x_3 + x_4 = 3 ;$$

$$\Rightarrow x_4 = -3x_3 + 3$$

$$\Rightarrow x_1 + 3x_2 - 3x_3 = -5$$

$$\text{let } x_1 = s, \quad x_3 = t$$

$$\text{solution} = \begin{bmatrix} s \\ (-5-s+3t)/3 \\ t \\ 3-3t \end{bmatrix}$$

7c)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Augmented Matrix $\left\{ \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{array} \right.$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -3 & -6 & -3 \\ 0 & -6 & -12 & -6 \end{array} \right] \rightarrow R_3 \rightarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -3 & -6 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow R_2 \rightarrow R_2 / (-3)$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow R_1 \rightarrow R_1 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$x_1 - x_3 = -1$$

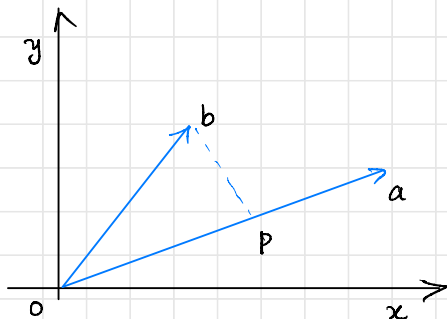
$$x_2 + 2x_3 = 1$$

$$\text{let } x_3 = m$$

$$x_1 = m - 1, \quad x_2 = 1 - 2m$$

$$\text{solution: } \begin{bmatrix} m-1 \\ 1-2m \\ m \end{bmatrix}$$

9)



$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

(Dot product)

$$\therefore \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$\Rightarrow \frac{a_1 b_1}{\sqrt{a_1^2 + a_2^2}} + \frac{a_2 b_2}{\sqrt{b_1^2 + b_2^2}}$$

$$\theta = \cos^{-1} \left[\frac{a_1 b_1}{\sqrt{a_1^2 + a_2^2}} + \frac{a_2 b_2}{\sqrt{b_1^2 + b_2^2}} \right]$$

$$\cos \theta = \frac{OP}{|\vec{b}|}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{OP}{|\vec{b}|} \Rightarrow OP = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

$$\begin{aligned} \Rightarrow \vec{OP} &= \frac{\vec{a}}{|\vec{a}|} \cdot \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \\ &= \frac{\vec{a} \cdot \vec{a} \cdot \vec{b}}{|\vec{a}|^2} \end{aligned}$$

10.a)

$$x = (x_1, x_2, x_3, x_4)$$

$$\text{given, } x_1 = x_2 = x_3$$

$$\therefore x = (x_1, x_1, x_1, x_4)$$

$$= (x_1, x_1, x_1, 0) + (0, 0, 0, x_4)$$

$$= x_1 (1, 1, 1, 0) + x_4 (0, 0, 0, 1)$$

$(1, 1, 1, 0)$ and $(0, 0, 0, 1)$ span the vector space and form a basis since they are linearly independent.

10.b)

Dimension of a subspace is given by the number of vectors in a basis in the subspace. For the given set of vectors $x_1, x_2, x_3, \dots, x_m$ we can't be very sure that all the vectors are linearly independent. Therefore, $\dim(S) \leq m$ not always equal to m .

11.)

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & x_1 \\ 1/\sqrt{3} & 0 & x_2 \\ 1/\sqrt{3} & -1/\sqrt{2} & x_3 \end{bmatrix}$$

A matrix will be orthogonal only if $Q \cdot Q^T = I$.

$$Q^T = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ x_1 & x_2 & x_3 \end{bmatrix}$$

$$Q \cdot Q^T = I$$

$$\begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & x_1 \\ 1/\sqrt{3} & 0 & x_2 \\ 1/\sqrt{3} & -1/\sqrt{2} & x_3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5/6 + x_1^2 & 1/3 + x_1 x_2 & -1/6 + x_1 x_3 \\ 1/3 + x_1 x_2 & 1/3 + x_2^2 & 1/3 + x_2 x_3 \\ -1/6 + x_1 x_3 & 1/3 + x_2 x_3 & 5/6 + x_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{lll} 5/6 + x_1^2 = 1 & 1/3 + x_1 x_2 = 0 & -1/6 + x_1 x_3 = 0 \\ 1/3 + x_1 x_2 = 0 & 1/3 + x_2^2 = 1 & 1/3 + x_2 x_3 = 0 \\ -1/6 + x_1 x_3 = 0 & 1/3 + x_2 x_3 = 0 & 5/6 + x_3^2 = 1 \end{array}$$

upon solving, $x_1 = \pm 1/\sqrt{6}$, $x_2 = \pm \sqrt{2/3}$, $x_3 = \pm 1/\sqrt{6}$

Therefore,

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & \sqrt{2}/3 \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

$$\text{OR } Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & \sqrt{2}/3 \\ 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

12)

$$u = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$u^T = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$u \cdot u^T = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \Rightarrow \frac{1}{3} (I)$$

$$Q = I - 2uu^T$$

$$= \begin{bmatrix} 1/3 & -2/3 & -2/3 \\ -2/3 & 1/3 & -2/3 \\ -2/3 & -2/3 & 1/3 \end{bmatrix}$$

Orthogonal matrix $\rightarrow Q \cdot Q^T = I$

$$\Rightarrow \begin{bmatrix} 1/3 & -2/3 & -2/3 \\ -2/3 & 1/3 & -2/3 \\ -2/3 & -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1/3 & -2/3 & -2/3 \\ -2/3 & 1/3 & -2/3 \\ -2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore, $Q = I - 2uu^T$ is an orthogonal matrix.

6)

$$A = \begin{bmatrix} 0 & 1 & 1 & 2 & -1 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{bmatrix}$$

last three columns are linearly dependent on first 2 columns.

Rank of $A = 2$.

$$Ax = x_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$= (x_1 + x_2 + x_5) \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + (x_1 + x_3 + 2x_4 - x_5) \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 0$$

As the above vectors are linearly independent,

$$Ax = 0 \text{ if } x_1 + x_2 + x_5 = 0 \text{ and } x_1 + x_3 + 2x_4 - x_5 = 0.$$

let $x_3 = 1, x_4 = 0, x_5 = 0 \rightarrow$ then, $x_1 = -1$ and $x_2 = -1$

let $x_3 = 0, x_4 = 1, x_5 = 0 \rightarrow$ then, $x_1 = 0, x_2 = -2$

let $x_3 = 0, x_4 = 0, x_5 = 1 \rightarrow$ then, $x_1 = -1, x_2 = 1$

Basis for Null space, $\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

if $Ax = b$, as $\text{rank} = 2$. b should be in range space of A .

8) a)

$$x_1 + x_2 + x_3 + x_4 + x_5 = 1 \quad \text{--- ①}$$

$$x_2 = x_1/4 \quad x_1 - 4x_2 = 0 \quad \text{--- ②}$$

$$x_3 + x_4 = x_5 \quad x_3 + x_4 - x_5 = 0 \quad \text{--- ③}$$

So,

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & -4 & 0 \\ 0 & 0 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$x^* = A^T(AA^T)^{-1}B$$

$$x^* = \begin{bmatrix} 0.2581 \\ 0.2258 \\ 0.1935 \\ 0.0645 \\ 0.2581 \end{bmatrix}$$

⑥

parabola passes through

$(1, -2), (2, -1), (4, 0), (5, 1), (8, 2)$

$$y = Ex^2 + Dx + C.$$

substitute values: $E + D + C = -2$

$$4E + 2D + C = -1$$

$$16E + 4D + C = 0$$

$$25E + 5D + C = 1$$

$$64E + 8D + C = 2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 16 & 4 & 1 \\ 25 & 5 & 1 \\ 64 & 8 & 1 \end{bmatrix} \begin{bmatrix} E \\ D \\ C \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

$$x^* = (A^T A)^{-1} A^T b$$

$$= \begin{bmatrix} -0.0417 \\ 0.9417 \\ -2.85 \end{bmatrix} \Rightarrow \begin{bmatrix} E \\ D \\ C \end{bmatrix}$$

Therefore, $y = -0.0417x^2 + 0.9417x - 2.8500.$